United States Patent	12409299
Kind Code	B2
Date of Patent	September 09, 2025
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Bifurcated hub

Abstract

There is provided an introducer system for percutaneously delivering a first and a second medical device to a patient. The system may comprise an introducer sheath having a longitudinal axis and a lumen formed therein. The system may also comprise a bifurcated hub coupled to a proximal end of the sheath. The hub may comprise a first arm having a first lumen and a first hemostasis valve for the passage of the first medical device. The hub may also comprise a second arm coupled to the first arm and having a second lumen and a second hemostasis valve for the passage of the second medical device. Further, the hub may comprise a connection port such that the first lumen and the second lumen are in communication with the lumen of the introducer sheath for the passage of at least one of the first and second medical devices through the introducer sheath for delivery to the patient.

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Appl. No.: **17/024514**

Filed: **September 17, 2020**

Prior Publication Data

Document Identifier	Publication Date
US 20210085923 A1	Mar. 25, 2021

Related U.S. Application Data

us-provisional-application US 62903402 20190920

Publication Classification

Int. Cl.: **A61M25/00** (20060101); **A61M1/16** (20060101); **A61M39/06** (20060101); A61M60/422 (20210101)

U.S. Cl.:

CPC **A61M25/0097** (20130101); **A61M39/06** (20130101); A61M1/1698 (20130101); A61M2039/062 (20130101); A61M60/422 (20210101)

Field of Classification Search

CPC: A61M (25/0097); A61M (39/06); A61M (1/1698); A61M (60/422); A61M (2039/062); A61M (60/109); A61M (60/13); A61M (60/139); A61M (60/216); A61M (60/295); A61M (60/38); A61M (60/865); A61M (1/3653); A61M (29/00); A61M (2025/0681); A61M (2039/0626); A61M (39/0606); A61M (25/0014)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to U.S. Provisional Application No. 62/903,402 filed Sep. 20, 2019, the contents of which are fully incorporated herein by reference. (2) The present application is related to currently pending U.S. patent application Ser. No. 16/815,690 and International Application No. PCT/US20/22021, which were filed Mar. 11, 2020, and which claim priority to U.S. Provisional Application No. 62/817,901, filed Mar. 13, 2019, the contents of each of which are fully incorporated herein by reference.

BACKGROUND

(1) Currently, percutaneous mechanical support devices are leveraged for a variety of clinical indications. Such support devices may comprise, but are not limited to, an Impella® pump, an Extracorporeal Membrane Oxygenation (ECMO) pump, and a balloon pump. The Impella® pump may further comprise an Impella 2.5® pump, an Impella 5.0® pump, an Impella CP® pump and an Impella LD® pump, all of which are by Abiomed, Inc. of Danvers, MA. Most often they are inserted into a patient percutaneously through a single access point (e.g., radial access, femoral access, axillary access) while other procedures, such as, for example, percutaneous coronary intervention (PCI) are performed through a second access point, such as a contralateral femoral or radial access point. The use of multiple devices on a patient at the same time therefore often requires multiple access sites which presents several challenges.

BRIEF SUMMARY

(2) The present technology relates to systems and methods for percutaneously delivering a first medical device and a second medical device to a patient.

(3) In one aspect, the disclosure describes an introducer system comprising: an introducer sheath having a longitudinal axis and a lumen formed therein; and a hub coupled to a proximal end of the introducer sheath. The hub comprises: a first arm having a first lumen and a first hemostasis valve, the first lumen and the first hemostasis valve configured for the passage of a first medical device; a second arm coupled to the first arm and having a second lumen and a second hemostasis valve, the second lumen and the second hemostasis valve configured for the passage of a second medical device; and a connection port coupled to the introducer sheath and to the first arm and the second arm, such that the first lumen and the second lumen are in communication with the lumen of the introducer sheath to allow the passage of at least one of the first medical device and the second medical device through the introducer sheath for delivery to a patient. In some aspects, the first arm is arranged parallel to the longitudinal axis of the introducer sheath. In some aspects, the second arm is configured to branch off the first arm at an angle of no more than 90°. In some aspects, the first arm and the second arm are arranged in a Y-shaped configuration with respect to the introducer sheath. In some aspects, the second arm is located proximal to the connection port. In some aspects, the first arm and the second arm each has a proximal end and a distal end, and the distal end of the first arm is positioned distal of the proximal end of the second arm. In some aspects, the first hemostasis valve has a first opening and the second hemostasis valve has a second opening, and the first opening is smaller than the second opening. In some aspects, the first opening has a diameter of about 8 Fr. In some aspects, the second opening has a diameter of about 14 Fr. In some aspects, the second lumen merges with the first lumen within the hub. In some aspects, the introducer sheath comprises a single lumen for the passage of the first and second medical devices. In some aspects, the first lumen and the second lumen are maintained as separate lumens within the hub. In some aspects, the first lumen and the second lumen are fabricated via injection molding. In some aspects, the introducer sheath comprises a dual lumen sheath such that the first lumen is in communication with one of the lumens of the dual lumen sheath, and the second lumen is in communication with the other lumen of the dual lumen sheath. In some aspects, the introducer sheath is an expandable sheath. In some aspects, the introducer sheath is a peel-away sheath. In addition, the hub may further comprise tabs to enable separation of the hub and the peel-away sheath. In some aspects, the first and second hemostasis valves are configured to seal the respective first and second lumens. In some aspects, the first and second hemostasis valves are each configured to be penetrable by the first or second medical device. In some aspects, the hub further comprises at least one suture ring. In some aspects, the first arm and the second arm each comprise at least one side-port. In addition, the side-port may comprise an irrigation port configured to be

supplied with an irrigation fluid. In some aspects, at least one of the first arm and the second arm comprises a locking mechanism configured to prevent axial movement of one or both of the first medical device and the second medical device within the introducer sheath after delivery to the patient. In some aspects, at least one of the first arm and the second arm comprises a locking mechanism configured to prevent axial movement of one or both of the first medical device and the second medical device within the introducer sheath after delivery to the patient. In some aspects, the locking mechanism comprises at least one of: a Tuohy-Borst adaptor, an inflatable balloon, and a locking lever arm. In some aspects, the locking mechanism is biased in a state that is configured to prevent axial movement of one or both of the first medical device and the second medical device within the introducer sheath. In some aspects, the introducer sheath comprises at least one of: a polyether block amide; a polyethylene material; a polytetrafluoroethylene (PTFE) material; a high-density polyethylene (HDPE) material; a medium-density polyethylene (MDPE) material; or a low-density polyethylene (LDPE) material. In some aspects, the hub comprises at least one of: ethylene-vinyl acetate (EVA); styrene-butadiene copolymer (SBC); styrene ethylene butylene styrene (SEBS); a high-density polyethylene (HDPE) material; a medium-density polyethylene (MDPE) material; a low-density polyethylene (LDPE) material; polyether ether ketone (PEEK); a polyether block amide; an elastomer; synthetic rubber; or a polyethylene, polyurethane, or polycarbonate material with an elastic modulus of about 40 ksi. In some aspects, the first medical device is a mechanical circulatory support device, and the second medical device is a coronary reperfusion therapy device for providing the patient with percutaneous coronary intervention (PCI). In some aspects, the coronary reperfusion therapy device is a stent. In some aspects, the stent is configured for insertion through the second arm and introducer sheath by a catheter. In some aspects, the mechanical circulatory support device comprises at least one of: a blood pump; a transvalvular axial-flow (TV)-pump; an intra-aortic balloon pump; or an extracorporeal membrane oxygenation (ECMO) pump. In some aspects, the mechanical circulatory support device is a rotary blood pump having a cannula and a rotor and rotor housing. In some aspects, the first arm and the introducer sheath are configured to allow passage of the cannula of the rotary blood pump. In some aspects, the first arm and the introducer sheath are configured to allow passage of the rotor and rotor housing of the rotary blood pump. In some aspects, the hub comprises up to five second arms, each second arm configured with a hemostasis valve and a lumen in communication with the introducer sheath for the passage of the second medical device from a respective second arm into the introducer sheath. In some aspects, the second arms are arranged in a radially symmetric manner about the first arm. In some aspects, the hub comprises two second arms. In some aspects, the introducer system may further comprise at least one third arm coupled to the first arm, each third arm having a third lumen and a third hemostasis valve, the third lumen and third hemostasis valve configured for the passage of a third medical device. In some aspects, the connection port is configured to couple to a proximal end of the introducer sheath for the delivery of at least one of the first medical device or the second medical device to the patient. In some aspects, the introducer sheath comprises a valve positioned at a proximal end of the introducer sheath, and wherein the connection port is configured to penetrate the valve so as to enable the connection port to couple to the introducer sheath. In some aspects, the connection port comprises a connector to secure the proximal end of the introducer sheath to the hub. In some aspects, the connector comprises any one of a screw connector, a snap-fit connector, or an interference-fit connector.

(4) In another aspect, the disclosure describes a method comprising: inserting a first medical device into a first arm of an introducer hub, the first arm having a first lumen for the passage of the first medical device therethrough; inserting a second medical device into a second arm attached to the first arm, the second arm having a second lumen for the passage of the second medical device therethrough; providing the first medical device and the second medical device to an introducer sheath via a connection port of an introducer hub, the connection port coupled to a proximal end of the introducer sheath; and delivering the first medical device and the second medical device to a

patient from a distal end of the introducer sheath. In some aspects, the method further comprises inserting the first and second medical devices into a lumen formed within the introducer sheath for delivery to the patient. In some aspects, the method further comprises inserting the first medical device into a first lumen formed within the introducer sheath for delivery to the patient, and inserting the second medical device into a second lumen formed within the introducer sheath for delivery to the patient, the first lumen isolated from the second lumen. In some aspects, the method further comprises attaching the introducer hub to the patient via a suture ring. In some aspects, the method further comprises providing one or both of the first lumen and the second lumen with an irrigation fluid via a side-port positioned on each of the first and second arms. In some aspects, the method further comprises activating a locking mechanism to prevent axial movement of one or both of the first medical device and the second medical device within the introducer sheath. In some aspects, the locking mechanism comprises at least one of: a Tuohy-Borst adaptor, an inflatable balloon, and a locking lever arm. In some aspects, the locking mechanism is biased in a state that prevents axial movement of one or both of the first medical device and the second medical device within the introducer sheath. In some aspects, the method further comprises inserting a third medical device into a third arm attached to the first arm, the third arm having a third lumen for the passage of the third medical device therethrough for delivery to the patient. In some aspects, the method further comprises coupling the connection port to a proximal end of an introducer sheath for delivery of at least one of the first medical device or the second medical device. In some aspects, the method further comprises inserting the connection port through a valve positioned at the proximal end of the introducer sheath so as to enable the coupling of the connection port to the introducer sheath. In some aspects, the connection port comprises a connector to secure the proximal end of the introducer sheath to the hub. In some aspects, the connector comprises any one of a screw connector, a snap-fit connector, or an interference-fit connector. In some aspects, the method further comprises supporting the patient's heart that has sustained myocardial infarction. In some aspects, the method further comprises supporting the patient's heart that has sustained myocardial infarction. In some aspects, the method further comprises: inserting the first medical device through the first arm and through the introducer sheath into the patient's left ventricle; operating the first medical device for a support period of greater than 30 minutes at a rate of at least 2.5 L/min of blood flow; inserting the second medical device through the second arm and through the introducer sheath into a coronary vessel of the patient; and operating the second medical device after the support period has elapsed. In some aspects, the first device comprises a mechanical circulatory support device. In some aspects, the mechanical circulatory support device comprises at least one of: a blood pump, a transvalvular axial-flow (TV)-pump, an intra-aortic balloon pump, or an extracorporeal membrane oxygenation (ECMO) pump. In some aspects, the second device comprises a coronary reperfusion therapy device for providing the patient with percutaneous coronary intervention (PCI). In some aspects, the mechanical circulatory support device is operated to pump blood from the patient's left ventricle into the patient's aorta during the support period. In some aspects, the second medical device is inserted through the second arm after the first medical device is positioned across the patient's aortic valve and is unloading the patient's left ventricle. In some aspects, the second medical device is inserted through the introducer sheath at least 15 minutes after the first medical device begins unloading the patient's left ventricle. In some aspects, the first medical device is positioned with a distal tip located within the patient's left ventricle and pumps blood from the patient's left ventricle into the patient's aorta. In some aspects, the introducer hub comprises up to five second arms, each second arm configured with a hemostasis valve and a lumen in communication with the introducer sheath for the passage of the second medical device from a respective second arm into the introducer sheath. In some aspects, the second arms are arranged in a radially symmetric manner about the first arm. In some aspects, the introducer hub comprises two second arms. In some aspects, the introducer hub further comprises at least one third arm coupled to the first arm, each third arm having a third lumen and a third hemostasis valve, the third lumen and third hemostasis valve configured for the passage of a third medical device.

(5) In another aspect, the disclosure describes an introducer hub comprising: a first arm having a first lumen and a first hemostasis valve, the first lumen and first hemostasis valve configured for the passage of a first medical device; a second arm coupled to the first arm and having a second lumen and a second hemostasis valve, the second lumen and second hemostasis valve configured for the passage of a second medical device; and a connection port coupled to an introducer sheath and to the first arm and the second arm, such that the first lumen and the second lumen are in communication with the lumen of the introducer sheath to allow the passage of at least one of the first medical device and the second medical device through the introducer sheath for delivery to a patient. In some aspects, the first arm is arranged parallel to a longitudinal axis of the introducer sheath. In some aspects, the second arm is configured to branch off the first arm at an angle of no more than 90°. In some aspects, the first arm and the second arm are arranged in a Y-shaped configuration with respect to the introducer sheath. In some aspects, the second arm is located proximal to the connection port. In some aspects, the first arm and the second arm each has a proximal end and a distal end, and the distal end of the first arm is positioned distal of the proximal end of the second arm. In some aspects, the first hemostasis valve has a first opening and the second hemostasis valve has a second opening, and the first opening is smaller than the second opening. In some aspects, the first opening has a diameter of about 8 Fr. In some aspects, the second opening has a diameter of about 14 Fr. In some aspects, the second lumen merges with the first lumen. In some aspects, the introducer sheath comprises a single lumen for the passage of the first and second medical devices. In some aspects, the first lumen and the second lumen are maintained as separate lumens. In some aspects, the first lumen and the second lumen are fabricated via dual lumen extrusion. In some aspects, the introducer sheath comprises a dual lumen sheath such that the first lumen is in communication with one of the lumens of the dual lumen sheath, and the second lumen is in communication with the other lumen of the dual lumen sheath. In some aspects, the introducer sheath is an expandable sheath. In some aspects, the introducer sheath is a peel-away sheath. In some aspects, the introducer hub further comprises tabs to enable separation of the introducer hub and the peel-away sheath. In some aspects, the first and second hemostasis valves are configured to seal the respective first and second lumens. In some aspects, the first and second hemostasis valves are each configured to be penetrable by the first or second medical device. In some aspects, the introducer hub further comprises at least one suture ring. In some aspects, the first arm and the second arm each comprise at least one side-port. In some aspects, the side-port comprises an irrigation port configured to be supplied with an irrigation fluid. In some aspects, the hub comprises an indent configured to match the size of at least one of the first medical device or the second medical device to facilitate the passage of the first and second medical devices through the hub. In some aspects, at least one of the first arm and the second arm comprises a locking mechanism configured to prevent axial movement of one or both of the first medical device and the second medical device within the introducer sheath after delivery to the patient. In some aspects, the locking mechanism comprises at least one of: a Tuohy-Borst adaptor, an inflatable balloon, and a locking lever arm. In some aspects, the locking mechanism is biased in a state that is configured to prevent axial movement of one or both of the first medical device and the second medical device within the introducer sheath. In some aspects, the introducer hub comprises up to five second arms, each second arm configured with a hemostasis valve and a lumen in communication with the introducer sheath for the passage of the second medical device from a respective second arm into the introducer sheath. In some aspects, the second arms are arranged in a radially symmetric manner about the first arm. In some aspects, the introducer hub comprises two second arms. In some aspects, the introducer hub further comprises at least one third arm coupled to the first arm, each third arm having a third lumen and a third hemostasis valve, the third lumen and the third hemostasis valve configured for the passage of a third medical device. In

some aspects, the connection port is configured to couple to a proximal end of an introducer sheath for the delivery of at least one of the first medical device and the second medical device to a patient. In some aspects, the connection port is configured to penetrate a valve positioned at the proximal end of the introducer sheath so as to enable the connection port to couple to the introducer sheath. In some aspects, the connection port comprises a connector to secure the proximal end of the introducer sheath to the hub. In some aspects, the connector comprises any one of a screw connector, a snap-fit connector, or an interference-fit connector. In some aspects, the first medical device is a coronary reperfusion therapy device for providing the patient with percutaneous coronary intervention (PCI), and the second medical device is a mechanical circulatory support device. In some aspects, the coronary reperfusion therapy device is a stent. In some aspects, the mechanical circulatory support device is a blood pump, a transvalvular axial-flow (TV)-pump, an intra-aortic balloon pump, or an extracorporeal membrane oxygenation (ECMO) pump. In some aspects, the introducer hub comprises at least one of: ethylene-vinyl acetate (EVA); styrene-butadiene copolymer (SBC); styrene ethylene butylene styrene (SEBS); a high-density polyethylene (HDPE) material; a medium-density polyethylene (MDPE) material; a low-density polyethylene (LDPE) material; polyether ether ketone (PEEK); a polyether block amide; an elastomer; synthetic rubber; or a polyethylene, polyurethane, or polycarbonate material with an elastic modulus of about 40 ksi.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other objects and advantages will be apparent upon consideration of the following detailed description, taken in conjunction with the accompanying drawings, in which like reference characters refer to like parts throughout, and in which:

(2) FIG. **1** shows an illustrative cross section of a dual hub introducer sheath system used for delivering a first medical device and second medical device into an arteriotomy of a patient, according to aspects of the disclosure;

(3) FIG. **2** shows a dual hub introducer sheath system with a dilator for insertion into the arteriotomy of the patient, according to aspects of the disclosure;

(4) FIG. **3** shows a detailed view of the dual hub introducer sheath system of FIG. **2**;

(5) FIG. **4** shows a detailed view of the dual hub introducer sheath system of FIG. **2** with an occulder sealing a lumen in a side arm;

(6) FIG. **5**A shows an illustrative locking mechanism, in an open state, used in a dual hub introducer sheath system according to aspects of the disclosure;

(7) FIG. **5**B shows the locking mechanism of FIG. **5**A in a locked state;

(8) FIG. **6**A shows an illustrative cross section of the dual hub of FIGS. **2** and **3** taken at the connection port, according to aspects of the disclosure;

(9) FIG. **6**B shows an illustrative cross section of the dual hub of FIGS. **2** and **3** taken at the connection port, with a notch or indent formed in the lumen of the dual hub, according to aspects of the disclosure;

(10) FIG. **7** shows an illustrative assembly of the dual hub coupled to the hub of an introducer sheath, according to aspects of the disclosure;

(11) FIG. **8** shows an illustrative flowchart of a method of using a dual hub introducer sheath system according to aspects of the disclosure; and

(12) FIG. **9** shows an illustrative flowchart of a method of using a dual hub introducer sheath system for unloading the left ventricle of the heart according to aspects of the disclosure.

DETAILED DESCRIPTION

(13) To provide an overall understanding of the systems, devices and methods described herein,

certain illustrative examples will be described. Although the examples and features described herein are specifically described for use in connection with dual hub introducer sheath for use in intravascular procedures involving catheter based ventricular assist devices, it will be understood that all the components and other features outlined below may be combined with one another in any suitable manner and may be adapted and applied to other types of procedures requiring a dual hub introducer sheath.

(14) As mentioned above, while it is possible to use multiple devices on a patient at the same time using multiple access sites, this can be challenging for a variety of reasons. Firstly, the patient may not have two anatomically available access sites for the PCI procedural devices, in which, for example, two 6-9 Fr sheaths may be used to facilitate procedures such as ballooning and stenting. In addition, peripheral artery disease, vessel lumen size (too small), scar tissue from previous procedures, and other diseases may complicate gaining access to a percutaneous site for larger devices, e.g., mechanical support devices. Using multiple access sites may also increase the likelihood of encountering vascular access complications, which can correlate to increased mortality, added hospital costs, etc. Further, multiple access sites requires more procedural time since access needs to be gained more than once, and can lead to increased procedural costs due to requiring multiple vascular closure devices, additional introducers, etc. There is thus a significant need for reducing the complexity of procedures requiring the operation of multiple devices on a patient.

(15) The systems, devices and methods described herein relate to a dual hub introducer sheath which enables a single access site for multiple devices. For purposes of illustration herein, but not by way of limitation, the devices are described as a mechanical assist device (such as, for example, an Impella device) and a device for the PCI procedure. However, one skilled in the art will understand that the present disclosure is not limited to any particular kind of percutaneously inserted device. In fact, the disclosure contemplates that, in some aspects of the technology, the multiple devices can be two of the same device. Until recently, such single access for multiple devices has not been apparently possible because physicians were unaware of the ability to fit both the PCI device and the Impella device through a single sheath without increasing the overall diameter of the sheath.

(16) Successful insertion of PCI devices through the same access sheath as the Impella device to perform both PCI and Impella support in a single access site has recently been reported (M. L. Esposito et al., “Left Ventricular Unloading Before Reperfusion Promotes Functional Recovery After Acute Myocardial Infarction,” Journal of the American College of Cardiology, Elsevier, vol. 72, no. 5, May 2018). However, using the current solutions leads to issues with hemostasis from the introducer valve and pump migration into and out of the ventricle as PCI devices are exchanged and manipulated. These adverse effects arise because conventional introducer valves are simply not designed to be accessed by dual devices.

(17) The devices and methods described herein relate to a dual hub introducer sheath having a longitudinal axis and a lumen formed therein. The sheath also comprises a bifurcated hub coupled to a proximal end of the introducer sheath. The hub comprises a first arm having a first lumen and a first hemostasis valve, the first lumen and hemostasis valve configured for the passage of a first medical device. The hub also comprises a second arm coupled to the first arm and having a second lumen and a second hemostasis valve, the second lumen and hemostasis valve configured for the passage of a second medical device. Further, the hub comprises a connection port coupled to the introducer sheath and to the first arm and the second arm, such that the first lumen and the second lumen are in communication with the lumen of the introducer sheath to allow the passage of at least one of the first medical device and second medical device through the introducer sheath for delivery to the patient.

(18) The dual hub introducer sheath of the present disclosure allows for both the PCI and Impella device to be inserted through it while maintaining appropriate and acceptable hemostasis. By

leveraging a bifurcated hub, two separate valves can be implemented which are specifically designed to meet insertion force and leakage requirements for either the Impella device or the PCI device, noting that these requirements and designs are quite different. Additionally, the dual hub introducer sheath of the present disclosure has a locking mechanism isolated to an arm of the hub intended for the Impella device which the physician can activate to hold the Impella in place preventing it from advancing or retracting as the PCI procedure is performed.

(19) FIG. **1** shows a dual hub introducer sheath delivery system **100** for percutaneously delivering a first medical device and a second medical device to a patient. System **100** comprises introducer sheath **110** that extends between a proximal end **112** and a distal end (not shown) along a longitudinal axis (not shown). Sheath **110** further comprises a lumen **115** that extends between the proximal end **112** and distal end for the passage of the first medical device and the second medical device. While FIG. **1** depicts a sheath **110** having a single lumen **115**, in some aspects of the technology, sheath **110** may comprise two distinct lumens throughout the length of the sheath, for example. In other aspects of the technology, sheath **110** may comprise any number of distinct lumens. System **100** further comprises a hub **120** coupled to the proximal end **112** of the sheath **110**.

(20) Hub **120** comprises a first arm **130** having a proximal end **132** and a distal end **134**, the first arm **130** defining a first lumen **135**. A first valve **138** is provided at the proximal end **132** of the first arm **130** to seal the first lumen **135** from the ambient. The first valve **138** is penetrable by the first medical device **140** and provides a first opening for passage of the first medical device **140**. Hub **120** further comprises a second arm **150** attached to the first arm **130**. As with the first arm **130**, second arm **150** defines a second lumen **155** and comprises a proximal end **152** and a distal end **154**. A second valve **158** is provided at the proximal end **152** of the second arm **150** to seal the second lumen **155** from the ambient. The second valve **158** is penetrable by the second medical device **160** and provides a second opening for passage of the second medical device **160**. In some aspects of the technology, the first valve **138** and the second valve **158** may comprise hemostasis valves (also referred to as “hemostatic” valves), such as, for example the valve described in U.S. Pat. No. 10,576,258 entitled “Hemostatic Valve for Medical Device Introducer,” the entire contents of which are hereby incorporated by reference herein. While FIG. **1** shows a hub **120** comprising one second arm **150**, it will be understood that the hub **120** may comprise any number of second arms arranged relative to the first arm **130**. In some aspects of the technology, the first opening may be smaller than the second opening. Likewise, in some aspects of the technology, the first opening may be larger than the second opening. This may be done to prevent or aid in preventing the user from inserting an incorrect device into the respective valves. For example, the first opening may have a diameter of about 8 Fr and the second opening may have a diameter of about 14 Fr. The first and second openings may be configured in this way, for example, to prevent a medical device having a diameter of about 14 Fr from being inserted into the first valve.

(21) The hub **120** further comprises a connection port **170** which connects to the first lumen **135** and the second lumen **155**. The connection port **170** of the hub **120** is coupled to the proximal end **112** of the sheath **110** such that the first medical device **140** and the second medical device **160** may traverse the sheath **110** to be delivered to a patient when the sheath **110** is inserted in the patient. In some aspects of the technology, such coupling may be a friction fit, for example, in which the proximal end **112** of the sheath **110** is dimensioned such that a friction fit between the outer surface of the sheath **110** and the inner surface of the connection port **170** prevents the proximal end **112** of the sheath **110** from detaching from the hub **120**. In other aspects of the technology, the coupling may be brought about by an external thread on the outer surface of the proximal end **112** of the sheath **110** which interacts with a complementary thread on the inner surface of the connection port **170**, for example. In further aspects of the technology. the sheath **110** may be coupled to the connection port **170** in any manner that enables the first lumen **135** and the second lumen **155** to be fluidically connected to the lumen **115** of the sheath **110** via the connection port **170** of the hub **120**.

In some aspects of the technology, the hub **120** may be overmolded and press fit or compressed onto the proximal end **112** of the sheath **110**.
(22) In some aspects of the technology, the hub **120** may be fabricated such that the first lumen **135** and the second lumen **155** merge within the hub **120** before transitioning into the connection port **170**, as shown in FIG. **1**. In such cases, the hub **120** may be coupled to a single lumen sheath, such as sheath **110** in FIG. **1**, where the lumen **115** of the sheath **110** is in fluid communication with the first lumen **135** and the second lumen **155** via the connection port **170** of the hub **120**. Thus, when the first medical device **140** is inserted into the first arm **130** and the second medical device **160** is inserted into the second arm **150** of the hub **120**, the medical devices share the same lumen **115** of the sheath **110** when traversing the sheath **110** for delivery to the patient. In other aspects of the technology, the hub **120** may be fabricated such that the first lumen **135** and the second lumen **155** are maintained as separate lumens throughout the hub **120**. Such fabrication may include formation of dual lumens within the hub by injection molding. In such cases, the hub **120** may be coupled to a dual lumen sheath such that the first medical device **140** and the second medical device **160** are separated at all times while traversing the length of the sheath **110** for delivery into an arteriotomy of the patient.
(23) As shown in FIG. **1** the second arm **150** may be arranged such that it branches off the first arm **130** at an angle relative to the first arm **130**, and the first arm **130** may be axially aligned with the longitudinal axis of the sheath **110**. In some aspects of the technology, this angle is not larger than 90°. In such cases, when the first medical device **140** is inserted into the first arm **130**, it is maintained in a substantially straight shape within the hub **120** without having to bend or kink; and when the second device **160** is inserted into the second arm **150**, the arrangement of the second arm **150** relative to the first arm **130** causes the second device **160** to bend such that it aligns with the first device **140** before exiting the hub **120** via the connection port **170** and traversing the lumen **115** of the sheath **110**.
(24) In some aspects of the technology, the first arm **130** and the second arm **150** may be arranged such that they form a Y-shaped configuration with respect to the longitudinal axis of the introducer sheath **110**. In such cases, both the first medical device **140** and the second medical device **160** may bend within the hub **120** such that they are aligned with the longitudinal axis of the sheath **110** as they exit the connection port **170** and traverse the lumen **115** of the sheath **110**.
(25) FIG. **2** shows the configuration of a dual hub introducer sheath system **200** according to aspects of the technology. Sheath system **200** is similar to sheath system **100** in that it comprises a sheath **210** having a proximal end **212** and a distal end **214** with a lumen extending between the proximal end **212** and the distal end **214** for the passage of at least one medical device, such as the first medical device **140** and the second medical device **160** of FIG. **1**, for delivery to an arteriotomy of a patient. The proximal end **212** of the sheath **210** is coupled to a connection port **270** of a hub **220**. The hub **220** comprises a first arm **230** having a proximal end **232** and a distal end **234**, and a second arm **250** attached to the first arm **230** and having a proximal end **252** and a distal end **254**. The first arm **230** forms a first lumen **235** which is sealed from the ambient by a first valve **238**. Similarly, the second arm **250** forms a second lumen **255** which is sealed from the ambient by a second valve **258**. The first valve **238** and the second valve **258** may comprise a hemostasis valve, and may be penetrable by the first and second medical devices, as has been described in relation to FIG. **1**. The first valve **238** provides a first opening for passage of the first medical device **140**, and the second valve **258** provides a second opening for passage of the second medical device **160**. As already noted, in some aspects of the technology, the first opening may be smaller than the second opening or vice versa. This may be done, for example, to prevent the user from inserting an incorrect device into the respective valves. For example, the first opening may have a diameter of about 8 Fr and the second opening may have gave a diameter of about 14 Fr so as to prevent a medical device having a diameter greater than 8 Fr from being inserted into the first valve. In some aspects of the technology, the second lumen **255** fluidically connects to the first

lumen **235** within the hub **220**, as shown in FIG. **2**. In other aspects of the technology, the first lumen **235** and the second lumen **255** may be maintained as separate lumens within the hub **220**. Again, such separated lumens may be fabricated by injection molding, for example. In some aspects of the technology, the first valve **238** and the second valve **258** may be inserted into position by a snap cap to secure their position within the hub **220**.
(26) As shown in FIG. **2** (and the enlargement **300** in FIGS. **3** and **4**) the first arm **230** further comprises a first side port **236** that is in fluid connection with the first lumen **235**. Similarly, the second arm **250** comprises a second side port **256** that is in fluid communication with the second lumen **255**. Side port **236** and side port **256** may each serve as an irrigation port through which irrigation fluid can be injected to free the lumens **235**, **255** within the hub **220** of any thrombus that may have formed during treatment of the patient. In other aspects of the technology, the side ports **236**, **256** may serve as inflation ports connected to inflatable balloons within the hub **220** that may be inflated with an inflation fluid to expand the balloons so as to anchor or lock the positions of the first medical device **140** and the second medical device **160** relative to the respective arms through which they are inserted. In such cases, the inflated balloon may compress the medical device against its respective arm so as to prevent axial movement of the medical device within the sheath once the device is deployed in the patient. Such locking mechanisms for securing the position of a first medical device to prevent axial motion during insertion or manipulation of a second medical device using internal sheath balloons are known to those skilled in the art. For example, various locking mechanisms for securing the position of a first medical device to prevent axial motion during insertion or manipulation of a second medical device using internal sheath balloons are described in U.S. Provisional Patent Application No. 62/797,527, the entire contents of which are hereby incorporated by reference herein. Other locking mechanisms will be detailed in the foregoing sections. Although various locking mechanisms may be described herein as securing a first medical device during insertion of a second medical device, it will be appreciated that all locking mechanisms described herein may also be used to secure the position of a second medical device to prevent axial motion of the second medical device during insertion or manipulation of a first medical device. While only one side port is shown on each arm in FIGS. **2** and **3**, any number of side ports may be used on each arm within the scope of the present disclosure.
(27) As discussed above, the second arm **250** may be arranged on the first arm **230** and configured to branch off the first arm **230** at an angle of no more than 90° with respect to the longitudinal axis of the sheath **210**. Additionally, in some aspects of the technology, the distal end **254** of the second arm **250** may be positioned proximal to the connection port **270**, and the proximal end **232** of the first arm **230** may be positioned distal to the connection port **270**. In this manner the proximal end **252** of the second arm **250** may be sufficiently spaced apart from the proximal end **232** of the first arm **230** to allow the first and second medical devices to interact with the respective arms **230**, **250** without having to abut each other.
(28) As seen in FIGS. **2** and **3**, the hub **220** may optionally include a suture ring **225** to aid with the attachment of the hub **220** to the patient after the first and second medical devices have been inserted in the patient. In certain aspects of the technology, the suture ring **225** may be positioned proximal to the connection port **270** as the profile of the hub **220** in this location may be smaller than at locations proximal to either the first arm **230** or the second arm **250**. In other aspects of the technology, the suture ring **225** may be located at any location along the body of hub **220**. While only one suture ring is shown in FIGS. **2** and **3**, it will be understood that any number of suture rings may be present to aid in securing the hub **220** to the patient.
(29) In order to insert the introducer sheath **210** into the patient, a dilator **280** may be used in connection with the dual hub **220**. FIGS. **2** and **3** also show a dilator **280** that has been inserted into the first arm **230** of the hub **220**. The dilator **280** comprises a proximal end **282** and a distal end **284**. The length of the dilator is such that the distal end **284** extends beyond the distal end **214** of the sheath **210** when the dilator **280** is fully inserted into the **210**. As previously mentioned, while

the first arm **230** and the second arm **250** may assume any configuration relative to the longitudinal axis of the sheath **210** (e.g., a Y-shaped configuration), where insertion of the introducer sheath **210** into the patient requires a dilator **280**, the first arm **230** may be axially aligned with the longitudinal axis of the sheath **210**. With this arrangement of the first arm **230**, the dilator does not need to bend when being inserted in to the hub **220**, which may allow for a greater force to be applied when inserting the sheath **210** into the patient. Once inserted, proximal end **282** of the dilator **280** may connect to the proximal end **232** of the first arm **230**. This may be accomplished by any suitable type of connection, such as through a press fit or a twist connection.

(30) In some aspects of the technology, an occluder **490** may be inserted into the lumen **255** of the second arm **250**, as shown in FIG. **4**. Such an occluder **490** may be inserted to prevent any backflow of fluid when the sheath **210** is inserted into the patient. This may be helpful in cases where sheath **210** needs to be repositioned after the medical devices have been removed from the first arm **230** and the second arm **250**. In such situations, due to their prior use, the respective seals **238**, **258** may not be able to seal the lumens **235**, **255** from the ambient completely due to wear and tear. As with the dilator **280**, the occluder **490** may connect to the proximal end **252** of the second arm **250** by any suitable type of connection, such as through a press fit or a twist connection.

(31) As mentioned in the foregoing, the first medical device **140** and the second medical device **160** may be axially constrained by a locking mechanism. In some aspects of the technology, a locking mechanism may be configured so that some action must be taken in order to lock and/or unlock it. In some aspects of the technology, a locking mechanism may have a bias. For example, in some aspects of the technology, a locking mechanism may be biased in an unlocked state such that it does not restrict movement of the medical device unless an action is taken to lock the locking mechanism. Conversely, in some aspects of the technology, a locking mechanism may be biased in a locked state such that it restricts movement of the medical device unless an action is taken to unlock the locking mechanism. In some aspects of the technology, the locking mechanism may comprise an internal balloon located within the first lumen **235** or the second lumen **255**, the internal balloon being inflated by a side arm **236**, **256**, as previously described. In some aspects of the technology, the locking mechanism may also comprise a locking lever arm as shown in FIGS. **5**A and **5**B. FIG. **5**A shows the cross section of a hub **510** similar to hubs **220** and **120** as described in the foregoing. Hub **510** is shown with a first medical device **520** traversing therethrough, although it will be appreciated that the hub **510** may be allow the passage of a plurality of medical devices. Hub **510** also comprises a lever arm **530** which may be a separate component which is positioned within the hub body. The lever arm **530** may be configured to have a semicircular shape as shown in FIGS. **5**A and **5**B, however any shape of arm may be used that is suitable to secure the medical device **520** and prevent axial motion thereof.

(32) The lever arm **530** may be pivotally connected to the hub **510** at a point **532** as shown in FIG. **5**A. In the unlocked position, the lever arm **530** resides within the hub body. The lever arm may comprise an actuating mechanism such as a handle or tab (not shown) that is accessible from the exterior of the hub **510**. The lever arm **530** comprises a notch or catch **538** that is configured to fit around the external periphery of the medical device **520** when the lever arm **530** is in the locked position. In this position, as shown in FIG. **5**B, the notch **538** pinches the medical device **520** to increase axial friction. In some aspects of the technology, the notch **538** may bend the medical device **520** when the lever arm **530** is in the locked position. In some aspects of the technology, the lever arm **530** may be located at the respective hemostasis valves **238**, **258**. Additionally, to secure the lever arm in the locked position, the end **534** of the lever arm **530** may be configured with a recess on its distal surface that engages with a protrusion **536** located within the hub body. The side profiles of end **534** and protrusion **536** are shown in FIG. **5**A. Similarly, FIG. **5**B shows end **534** in engagement with protrusion **536**. In some aspects of the technology, the lever arm **530** may be overmolded with a high friction material such as a low-durometer polyurethane or a silicone.

(33) In addition to, or as an alternative to, the locking mechanisms described in the foregoing, the

dual hub of the present disclosure may also comprise a Tuohy Borst mechanism built into the first or second arms of the hub body. Such a mechanism comprises a silicone slug that reduces in inner diameter onto the first and/or second medical device traversing the respective arm, thereby securing the position of the medical device.

(34) As noted above, the locking mechanisms described in the foregoing may be used in any context in which it is desired to secure a medical device. Accordingly, the locking mechanisms described above may be used to secure the position of the first medical device within the hub to prevent axial motion of the first medical device during insertion or manipulation of a second medical device. Likewise, the locking mechanisms described above may be used to secure the position of the second medical device within the hub to prevent axial motion of the second medical device during insertion or manipulation of a first medical device. Finally, the locking mechanisms described above may be included on multiple openings such that multiple medical devices can be secured relative to the hub.

(35) As noted above, in some aspects of the technology, the sheath **210** may comprise a dual lumen sheath. In such a case, when the dual lumen sheath is coupled to the hub **220**, the first lumen **235** of the hub **220** may be in fluid communication with one of the lumens of the dual lumen sheath, and the second lumen **255** of the hub **220** may be communication with the other lumen of the dual lumen sheath.

(36) In some aspects of the technology, the sheath **210** may comprise an expandable sheath. Expandable sheaths are well known to those skilled in the art and are not described in detail herein. For example, various expandable sheaths are described in U.S. Provisional Patent Application No. 62/797,527, which has been incorporated by reference herein.

(37) In some aspects of the technology, the sheath **210** may comprise a peel-away sheath. Peel-away sheaths are also well known to those skilled in the art and are not described in detail herein. For example, various peel-away sheaths are described in U.S. Provisional Patent Application No. 62/777,598, the entire contents of which are hereby incorporated by reference herein. Peel-away sheaths may comprise one or more lines of weakness that are formed within the sheath body and extend longitudinally along the sheath to allow the sheath to be pulled apart as needed during treatment of the patient.

(38) In some aspects of the technology, the hub **220** may comprise tabs that enable the hub **220** itself to be separated when it is no longer needed, e.g., when one or more of the medical devices are positioned within the patient.

(39) FIG. **6**A shows an exemplary cross-section **600** of the dual hub **220** of FIGS. **2** and **3**, according to aspects of the disclosure. The cross-section shown in FIG. **6**A is representative of a cross-section taken along any section of the hub **610** which allows for the passage of all the medical devices into the lumen of the introducer sheath. For example, FIG. **6**A may represent a section taken in close proximity to the connection port **270** of hub **220**, such as at line A-A′ in FIG. **3** which sees the passage of the first medical device and the second medical device into the introducer sheath **210**. In hub **610**, the first lumen of the first arm and the second lumen of the second arm merge into a single lumen towards the connection port (distal end) of the hub **610**.

(40) Also shown in FIG. **6**A is a first medical device **620** and a second medical device **630**. As previously described, the first medical device may be a coronary reperfusion therapy device for providing the patient with percutaneous coronary intervention (PCI), and the second medical device may be a mechanical circulatory support device, or vice versa. In some aspects of the technology, the coronary reperfusion therapy device may be a stent. In such cases, the stent may be configured for insertion through the second arm and introducer sheath by a catheter. In some aspects, the mechanical circulatory support device may be a blood pump, a transvalvular axial-flow (TV)-pump, an intra-aortic balloon pump, and an extracorporeal membrane oxygenation (ECMO) pump. In some aspects, the mechanical circulatory support device may be a rotary blood pump having a cannula, a rotor and rotor housing. In such cases, the rotary blood pump may be

configured such that the cannula can be inserted through the second arm of the introducer sheath. Likewise, the rotary blood pump may be configured such that the rotor and rotor housing can be inserted through the second arm and the introducer sheath.

(41) As seen in FIG. **6**A, the lumen **605** of the hub **610** at the connection port may be rigid such that it has a fixed diameter. In such cases, the diameter of the lumen **605** may therefore be a limiting factor when passing two or more medical devices through the hub **610** and into the introducer sheath. For example, in some cases, the second medical device **630** may be a mechanical circulatory device with a diameter of about 14 Fr, and the first medical device **620** may be a coronary reperfusion therapy device with a diameter of about 8 Fr. In such a case, if the lumen **605** of the hub **610** has a diameter of 14 Fr, there may be insufficient space within the lumen **605** of the hub **610** to fit both the coronary reperfusion therapy device (first medical device **620**) and the mechanical circulatory device (second medical device **630**). This scenario is depicted in FIG. **6**A, in which there is insufficient space for the first medical device **620** (shown with a broken line) once the second medical device **630** is inserted into the hub **610**.

(42) FIG. **6**B shows an exemplary cross section **650** of the dual hub **220** of FIGS. **2** and **3**, according to aspects of the disclosure. Dual hub **660** is similar to hub **610** of FIG. **6**A, but includes a notch or indent **670** formed in the hub body. Indent **670** may be sized to correspond to the diameter of the largest medical device to be inserted into hub **660**. As shown in FIG. **6**B, indent **670** is sized so as to accommodate at least a portion of the second medical device **630**, e.g., a mechanical circulatory device. The indent **670** therefore effectively increases the cross sectional area of the lumen **665** of the hub **660**, thereby enabling multiple devices to fit within the lumen **665** of the hub **660** as shown in FIG. **6**B, such as both a mechanical circulatory device (e.g., second medical device **630**) and a coronary reperfusion therapy device (e.g., first medical device **620**). In some aspects of the technology, the indent **670** may extend axially along the portion of the hub **660** through which multiple devices will pass, such as the distal portion of the hub **660** including the connection port (as depicted in FIGS. **1** and **2**). In some aspects, the indent **670** may extend axially throughout the hub **660** from the proximal end of one arm to the connection port.

(43) FIG. **7** shows a cross section **700** of an exemplary dual hub **710** similar to hubs **120** and **220** as described in the foregoing, according to aspects of the disclosure. Hub **710** comprises a first arm **720** having a first lumen **722** for the passage of a first medical device **725**, and a second arm **730** having a second lumen **732** for the passage of a second medical device **735**. While only one first arm **720** and one second arm **730** are shown in FIG. **7**, it will be understood that the hub **710** may include any number of first arms and second arms. In the example of FIG. **7**, the first lumen **722** is sealed from the atmosphere by a first valve **724**. Similarly, the second lumen **732** is sealed from the atmosphere by a second valve **734**. In some aspects of the technology, the first and/or second valves **724**, **734** may comprise hemostasis valves. As has been described above, the first valve **724** may provide a first opening for the passage of the first medical device **725** through the first lumen **722** of the first arm **720**, and the second valve **734** may provide a second opening for the passage of the second medical device **735** through the second lumen **732** of the second arm **730**. Here as well, in some aspects of the technology, the first and second openings may be sized differently (e.g., may have different diameters) to prevent penetration by an incorrect medical device. Additionally, while the first arm **720** and the second arm **730** are shown as arranged in Y-shaped configuration, in some aspects of the technology, the second arm **730** may be arranged as a side branch to the first arm **720** as shown in the examples of FIGS. **2-4**. Further, while the first lumen **722** and the second lumen **732** are shown as separate throughout the hub **710**, in some aspects of the technology, the first and second lumens **722** and **732** may merge to form a single lumen toward the distal end of the hub **710**. In the example of FIG. **7**, the distal end of the hub **710** comprises a connection port **740**.

(44) Here as well, the first medical device **725** may be a coronary reperfusion therapy device for providing the patient with percutaneous coronary intervention (PCI), and the second medical device **735** may be a mechanical circulatory support device. In some aspects of the technology, the

coronary reperfusion therapy device may be a stent. In such cases, the stent may be configured for insertion through the first arm **720** and introducer sheath **760** by a catheter (not shown). In some aspects of the technology, the mechanical circulatory support device may be a blood pump, a transvalvular axial-flow (TV)-pump, an intra-aortic balloon pump, and an extracorporeal membrane oxygenation (ECMO) pump. In some aspects, the mechanical circulatory support device may be a rotary blood pump having a cannula, a rotor and rotor housing. In such cases, the rotary blood pump may be configured such that the cannula can be inserted through the second arm **730** and the introducer sheath **760**. Likewise, the rotary blood pump may be configured such that the rotor and rotor housing can be inserted through the second arm **730** and the introducer sheath **760**.
(45) In the example of FIG. **7**, the dual hub **710** is coupled to the hub **750** of introducer sheath **760**. In this example, the connection port **740** of the dual hub **710** is inserted into a valve **754** of the hub **750** of the introducer sheath **760**. Once inserted in this way, the connection port **740** resides in a lumen **752** of the hub **750**, thereby enabling the first lumen **722** and the second lumen **732** of the dual hub **710** to be in fluid communication with the lumen **752** of the introducer sheath **760**. In some aspects of the technology, the hub **750** of the introducer sheath **760** may be configured such that it may be broken or separated about a shear line (e.g., a line running along dashed line **755**) upon application of a separation force at the proximal end of the hub **750**. Here as well, the introducer sheath **760** may comprise a single lumen as shown in FIG. **7**, or, alternatively, the introducer sheath **760** may comprise dual (or multiple) lumens such that the passageway for each of the medical devices are maintained as separate throughout the hub **710** and introducer sheath **760**. In some aspects of the technology, a coupler **770** may be mounted onto the distal end of the dual hub **710** to facilitate coupling with the hub **750** of the introducer sheath. For example, coupler **770** may be configured to attach the dual hub **710** to the hub **750** of the introducer sheath **760** via any one of a snap-fit, an interference fit, or a screw thread. Integration of the dual hub **710** of the present disclosure with existing introducer sheaths (e.g., introducer sheaths having single-port hubs) as shown in the example of FIG. **7** allows such introducer sheaths to be converted for the use of multiple medical devices (such as the first and second medical devices as described in the foregoing) while maintaining the structural integrity of the introducer sheath, as opposed to more destructive methods such as puncturing the sheath along its length to facilitate the insertion of a second medical device.
(46) In some aspects of the technology, the introducer sheaths described herein (e.g., elements **110**, **210**, **760**) may be extruded and/or laminated. In some aspects of the technology, the introducer sheaths described herein (e.g., elements **110**, **210**, **760**) may comprise at least one of: a polyether block amide (such as PEBAX® or PebaSlix®); a polyethylene material; a polytetrafluoroethylene (PTFE) material; a high-density polyethylene (HDPE) material; a medium-density polyethylene (MDPE) material; or a low-density polyethylene (LDPE) material.
(47) Further, as described above, in some aspects of the technology, the hub (e.g., elements **120**, **220**, **510**, **610**, **710**) may be formed by overmolding, extrusion, lamination, or any combination thereof. In some aspects of the technology, the hubs described herein (e.g., elements **120**, **220**, **510**, **610**, **710**) may comprise at least one of: ethylene-vinyl acetate (EVA); styrene-butadiene copolymer (SBC); styrene ethylene butylene styrene (SEBS); a high-density polyethylene (HDPE) material; a medium-density polyethylene (MDPE) material; a low-density polyethylene (LDPE) material; polyether ether ketone (PEEK); a polyether block amide (such as PEBAX® or PebaSlix®); an elastomer; synthetic rubber; a polyethylene, polyurethane, or polycarbonate material with an elastic modulus of about 40 ksi; a crack-resistant material; or a material with a low coefficient of friction.
(48) As mentioned in the foregoing description, the dual hub introducer sheath of the present disclosure is designed to facilitate the traversal of catheter-based medical devices (such as the first medical device and the second medical device) within the lumen of an introducer sheath. In some aspects of the technology, the first medical device is a mechanical circulatory support device, and the second medical device is a coronary reperfusion therapy device for providing the patient with

percutaneous coronary intervention (PCI). These PCI procedures may involve the use of a coronary stent delivered into the distal left anterior descending artery (LAD). Examples of such coronary stents include, but are not limited to, the Promus PREMIER™ and the REBEL™ bare-metal Platinum Chromium Coronary Stents, and the SYNERGY™ Bioabsorbable Polymer Stent, all by Boston Scientific, Marlborough, MA. In some aspects of the technology, the mechanical circulatory support device may comprise a rotary blood pump having a cannula, a rotor, and rotor housing. Examples of such blood pumps include, but are not limited to, an Impella® pump, an Extracorporeal Membrane Oxygenation (ECMO) pump, and a balloon pump. The Impella® pump may further comprise an Impella 2.5® pump, an Impella 5.0® pump, an Impella® pump, or an Impella LD® pump, all of which are by Abiomed, Inc. of Danvers, MA.

(49) In some aspects of the technology, the first medical device and the second medical device may be used with the dual hub introducer sheath as described in the foregoing in procedures where PCI and percutaneous ventricular assist devices are used in unison, such as, for example, the method of left ventricular unloading in treating myocardial infarction as described in U.S. patent application Ser. No. 16/244,998, the entire contents of which are hereby incorporated by reference herein.

(50) FIG. **8** illustrates an exemplary method **800** of using a dual hub introducer sheath, such as any of the introducer sheaths as described in the foregoing description, according to aspects of the technology. The method **800** will be described in relation to the exemplary systems depicted in FIGS. **1-7** above. Prior to using the dual hub introducer sheath, the sheath **210** is positioned into the arteriotomy of the patient (not indicated in FIG. **8**). Prior to insertion of the sheath **210** into the patient, a connection port of the hub, such as connection port **270** of the hub **220** as described above, is coupled to a proximal end of an introducer sheath, such as end **212** of sheath **210** as described above. In some aspects of the technology, a dilator may be inserted into the lumen of the sheath before insertion into the patient, such as dilator **280** shown in FIGS. **2** and **3**. The dilator assists with positioning the sheath in regions of the patient's body which are difficult to penetrate with the sheath alone. Once inserted, the dilator is removed from the lumen of the sheath.

(51) In step **810**, the first medical device **140** is inserted into a first arm **230** of the dual hub **220**. Here, the first medical device **140** is pushed through the first hemostasis valve **238** and traverses the hub **220** towards the connection port **270**. As with the first medical device **140**, in step **820**, the second medical device **160** is pushed through the second hemostasis valve **258** in the second arm **250**, after which it also traverses hub **220** toward connection port **270**. In step **830**, the first and second medical devices are provided to the lumen of the sheath **210** via the connection port **270** of the hub **220**. In step **840**, the first and second medical devices are delivered into an arteriotomy of the patient by pushing the devices along the length of sheath **210** until they exit the distal end **214** of the sheath **210**.

(52) Once in position in the patient's arteriotomy, the medical devices can be used as desired to treat the patient. In some aspects of the technology, the dual hub introducer sheath may be used to unload the left ventricle of the patient, as shown in the exemplary method **900** of FIG. **9**. In such cases, the first medical device **140** may be a mechanical circulatory support device, and the second medical device **160** may be a coronary reperfusion therapy device for providing the patient with percutaneous coronary intervention (PCI). With respect to FIG. **9**, in step **910**, after the first medical device **140** has emerged from the distal end **214** of the sheath **210**, it is advanced into position in the left ventricle of the heart. The first medical device **140** may then be locked in position via a locking mechanism located on the first arm **230** of the hub **220**. As already discussed, such locking mechanisms may include any one of: a Tuohy-Borst adaptor, an inflatable balloon, and a locking lever arm. Additionally, the hub may comprise an indent that is shaped to match the catheter of the mechanical circulatory support device such that the catheter of the mechanical circulatory support device can be housed in the indent thereby making room within the hub for the passage of the coronary reperfusion therapy device. In step **920**, the mechanical circulatory device is operated within the left ventricle for a support period in excess of 30 minutes. In some aspects of

the technology, the mechanical circulatory device may be operated at 2.5 L/min of blood flow. In step **930**, the second medical device **160** is positioned into a coronary vessel of the patient. The second medical device **160** may then also be locked in position via a locking mechanism located on the second arm **250** of the hub **220** as previously discussed. In step **940**, after the support period has passed, reperfusion therapy is applied to the coronary vessel via a PCI device, as described above. The reperfusion therapy may be performed in parallel with operation of the mechanical circulatory device, or after operation of the mechanical circulatory device.

(53) In some aspects of the technology, the various steps discussed above with respect to methods **800** and **900** of FIGS. **8** and **9** may be performed in a different order and/or concurrently with each other. Furthermore, if desired, one or more of the above described steps may be optional or may be combined. Likewise, while the above description has assumed that the first medical device is a coronary reperfusion therapy device for providing the patient with percutaneous coronary intervention, and the second medical device is a mechanical circulatory support device, it will be understood that such definition is interchangeable and thus that the first medical device may instead comprise the mechanical circulatory support device, and the second medical device may comprise the coronary reperfusion therapy device.

(54) In the foregoing disclosure, it will be understood that the term “about” should be taken to mean±20% of the stated value.

(55) The foregoing description is merely intended to be illustrative of the principles of the technology. As such, the devices and methods described herein can be practiced by other than the described implementations, which are presented for purposes of illustration and not of limitation. It is to be understood that the systems, devices and methods disclosed herein, while described with respect to certain procedures, may be applied in any context where access to an arteriotomy of a patient is desired without creating multiple access sites in the vasculature of the patient. In addition, the disclosed features may be implemented in any combination or subcombination (including multiple dependent combinations and subcombinations) with one or more other features described herein. The various features described or illustrated above, including any components thereof, may also be combined or integrated into other systems. Finally, certain features may be omitted or not implemented without departing from the spirit of the technology.

Claims

1. An introducer system comprising: a unitary hub having a proximal end and a distal end, the distal end of the unitary hub configured to be coupled to a proximal end of an introducer sheath, the introducer sheath comprising a lumen extending between a distal end of the introducer sheath and the proximal end of the introducer sheath, the lumen configured to allow a passage of at least one of a first medical device and a second medical device through the introducer sheath for delivery to a patient, the unitary hub comprising: a first arm having a first lumen and a first hemostasis valve, the first lumen and the first hemostasis valve configured for a passage of the first medical device; a second arm coupled to the first arm and having a second lumen and a second hemostasis valve, the second lumen and the second hemostasis valve configured for a passage of the second medical device; and an introducer sheath hub coupled to the proximal end of the introducer sheath that receives the distal end of the unitary hub, wherein the first lumen and the second lumen do not merge in the distal end of the unitary hub, wherein the distal end of the unitary hub is secured to the introducer sheath hub by a coupler and a third valve such that the first lumen and the second lumen are in communication with the lumen of the introducer sheath; and wherein the first hemostasis valve has a first opening and the second hemostasis valve has a second opening, and the first opening is smaller than the second opening.
2. The introducer system of claim 1, wherein the first opening has a diameter of about 8 Fr.
3. The introducer system of claim 2, wherein the second opening has a diameter of about 14 Fr.

4. The introducer system of claim 1, wherein the introducer sheath hub is configured to couple to the proximal end of the introducer sheath for the delivery of at least one of the first medical device or the second medical device to the patient.
5. The introducer system of claim 4, wherein the introducer sheath hub penetrates the third valve when coupled to the introducer sheath.
6. The introducer system of claim 5, wherein the coupler of the introducer sheath hub secures the unitary hub in the proximal end of the introducer sheath.
7. The introducer system of claim 6, wherein the coupler of the introducer sheath hub comprises any one of a screw connector, a snap-fit connector, or an interference-fit connector.
8. The introducer system of claim 1, wherein the introducer sheath hub has a shear line.
9. A method comprising: inserting an introducer sheath into a patient, the introducer sheath comprising an introducer sheath hub, a valve and a coupler positioned at a proximal end of the introducer sheath and a lumen extending between a distal end of the introducer sheath and the proximal end of the introducer sheath, the lumen configured to allow passage of a first medical device and a second medical device through the introducer sheath for delivery to the patient; inserting a distal end of an introducer hub through the coupler and valve and into the introducer sheath hub at the proximal end of the introducer sheath, wherein the coupler couples the distal end of the introducer hub to the introducer sheath hub at the proximal end of the introducer sheath; securing the distal end of the introducer hub to the introducer sheath hub at the proximal end of the introducer sheath; inserting the first medical device into a first arm of the introducer hub, the first arm comprising a first lumen of the introducer hub for a passage of the first medical device therethrough; inserting the second medical device into a second arm attached to the first arm, the second arm comprising a second lumen of the introducer hub for a passage of the second medical device therethrough; providing the first medical device and the second medical device to the introducer sheath through the coupler, valve and introducer sheath hub; and delivering the first medical device and the second medical device to the patient from the distal end of the introducer sheath, wherein the first and second lumens of the introducer hub are maintained as separate lumens within and throughout the introducer hub.
10. The method of claim 9, comprising: inserting the first and second medical devices into the lumen of the introducer sheath for delivery to the patient.
11. The method of claim 9, comprising: attaching the introducer hub to the patient via a suture ring.
12. The method of claim 9, comprising: providing one or both of the first lumen of the introducer hub and the second lumen of the introducer hub with an irrigation fluid via a side-port positioned on each of the first and second arms.
13. The method of claim 9, comprising: activating a locking mechanism to prevent axial movement of one or both of the first medical device and the second medical device within the introducer sheath.
14. The method of claim 13, wherein the locking mechanism comprises at least one of: a Tuohy-Borst adaptor, an inflatable balloon, and a locking lever arm.
15. The method of claim 14, wherein the locking mechanism is biased in a state that prevents axial movement of one or both of the first medical device and the second medical device within the introducer sheath.
16. The method of claim 9, comprising: inserting a third medical device into a third arm attached to the first arm, the third arm having a third lumen of the introducer hub for a passage of the third medical device therethrough for delivery to the patient.
17. The method of claim 9, wherein securing the introducer hub to the introducer sheath comprises snapping a portion of the introducer hub together with a portion of the introducer sheath.
18. The method of claim 9, wherein securing the introducer hub to the introducer sheath hub comprises screwing a portion of the introducer hub together with a portion of the coupler of the introducer sheath, a snap-fit connector, or an interference-fit connector.

19. The method of claim 9, further comprising: after providing the first medical device and the second medical device to the introducer sheath, breaking the introducer sheath hub along a shear line in the introducer sheath hub.